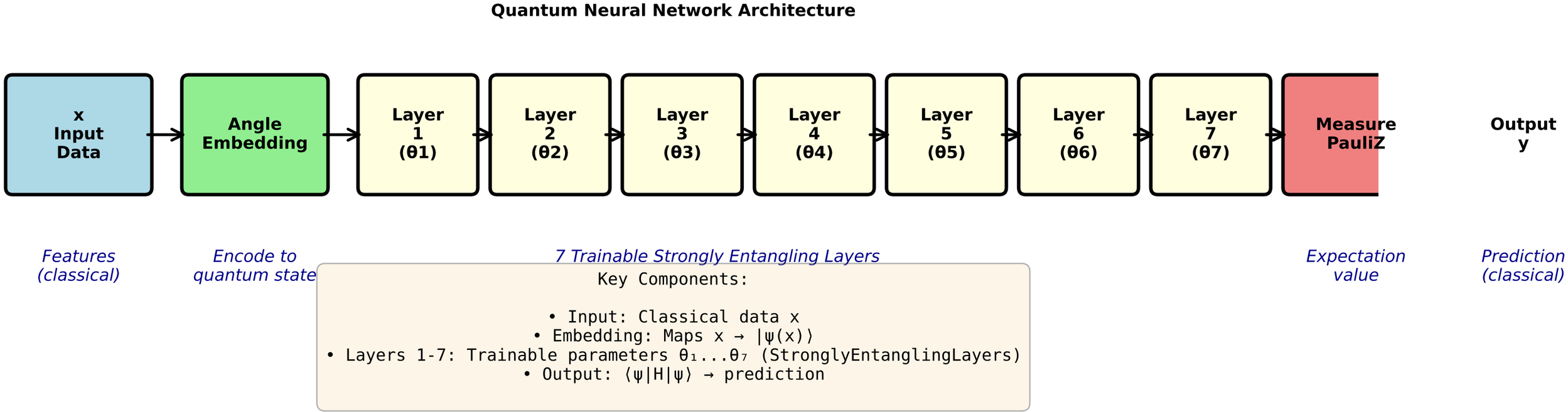
Quantum Neural Network Architecture
x
Input
Data
Angle
Embedding
Layer
1
(θ1)
Layer
2
(θ2)
Layer
3
(θ3)
Layer
4
(θ4)
Layer
5
(θ5)
Layer
6
(θ6)
Layer
7
(θ7)
Measure
PauliZ
Output
y
Features
(classical)
Encode to
quantum state
7 Trainable Strongly Entangling Layers
Expectation
value
Prediction
(classical)
Key Components:
• Input: Classical data x
• Embedding: Maps x → |ψ(x)⟩
• Layers 1-7: Trainable parameters θ₁...θ₇ (StronglyEntanglingLayers)
• Output: ⟨ψ|H|ψ⟩ → prediction